

Integration — Lesson 28 Test: Reading Accumulation Graphs

10 questions · show your work in the box · exact answers (leave π in them) · justifications must name $g' = f$ and its sign

Name: _____ Date: _____ Score: _____ / 10

GRAPH 1 (questions 1–7). The graph of f on $[0, 8]$ is made of straight line segments: it rises from $(0, 0)$ to $(2, 4)$, stays flat at height 4 from $t = 2$ to $t = 5$, then descends in a straight line from $(5, 4)$ to $(8, -2)$, crossing zero at $t = 7$. Sketch it lightly for yourself before starting. Let $g(x) = \int_0^x f(t) \, dt$.

1. Find $g(2)$. (What shape is the region?)

1 pt

2. Find $g(5)$.

1 pt

3. Find $g(7)$ and $g(8)$. (Careful: part of the last region is below the axis.)

1 pt

4. Find $g'(1)$ and $g'(6)$.

1 pt

5. On what interval is g increasing, and on what interval is g decreasing? Justify with a full sentence.

1 pt

6. At what x does g have a relative maximum? Justify with a full sentence, and give the value of g there.

1 pt

7. On what interval is g concave up, and on what interval is g concave down? Justify. (What is g doing on $(2, 5)$?)

1 pt

GRAPH 2 (questions 8–10). The graph of f on $[0, 10]$: a half-circle of radius 2 ABOVE the axis from $t = 0$ to $t = 4$ (centered at $(2, 0)$), then a straight line descending from $(4, 0)$ to $(6, -2)$, then flat at height -2 from $t = 6$ to $t = 10$. Let $g(x) = \int_0^x f(t) dt$.

8. Find $g(4)$, exactly. (What is the area of a half-circle of radius r ?)

1 pt

9. Find $g(6)$ and $g(10)$, exactly.

1 pt

10. (a) Find the exact value of $x > 0$ where $g(x) = 0$. (Hint: on the flat piece, g falls at a steady 2 per unit — write $g(x) = g(6) - 2(x - 6)$ and solve.) (b) If $h(x) = 1 + \int_0^x f(t) dt$, find $h(4)$, exactly.

1 pt
